

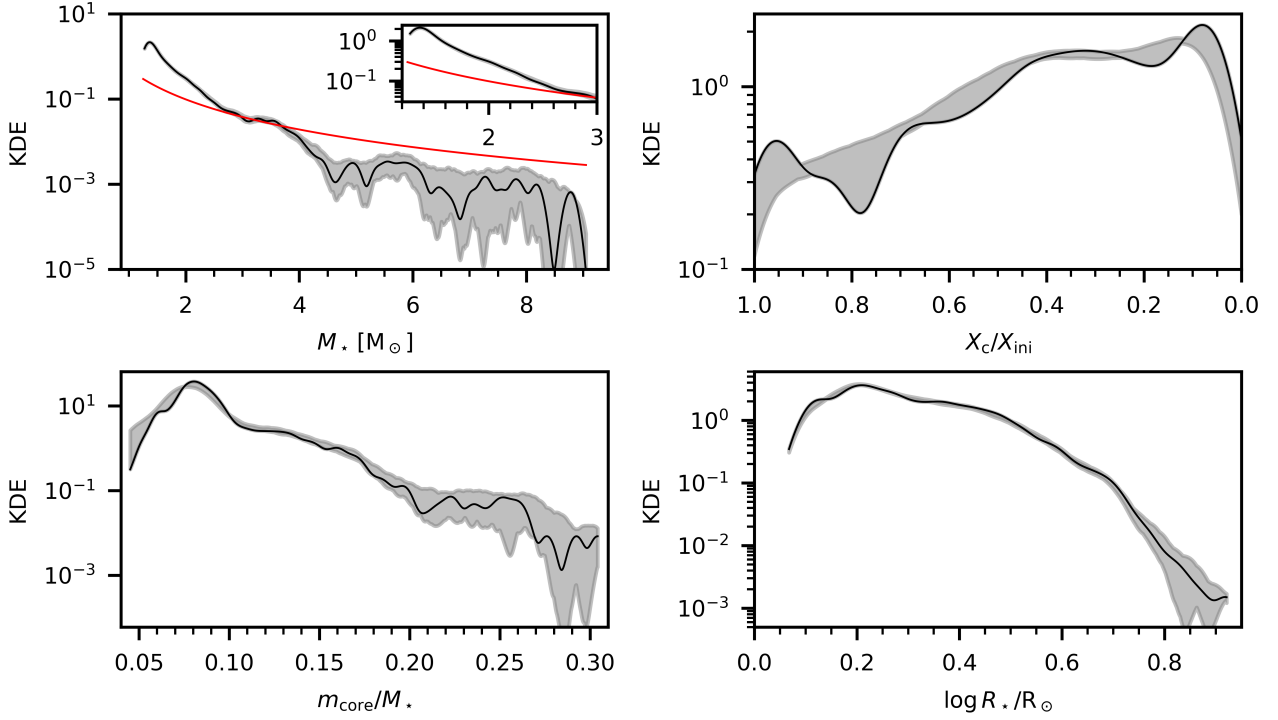


Fig. 4: Same as Fig. 3, but for $P_{\text{pred}} \geq 0.5$ (in black) and the result of 1000 iterations of bootstrapping (in grey). The inset in the top-left panel shows a zoom-in around the mass range of γ Dor pulsators.

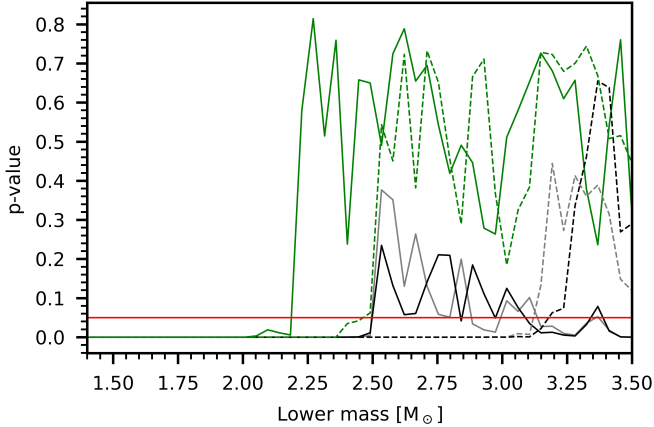


Fig. 5: Resulting p value from a one-sample Kolmogorov-Smirnov test between the inferred mass distribution of the sample of g-mode pulsators and a Salpeter IMF, as a function of the mass range tested. The upper mass is always $4 M_{\odot}$. The solid lines correspond to $Z = 0.0045$, the dashed lines correspond to $Z = 0.014$. The solid red line indicates a p value of 0.05 below which we reject the null hypothesis.

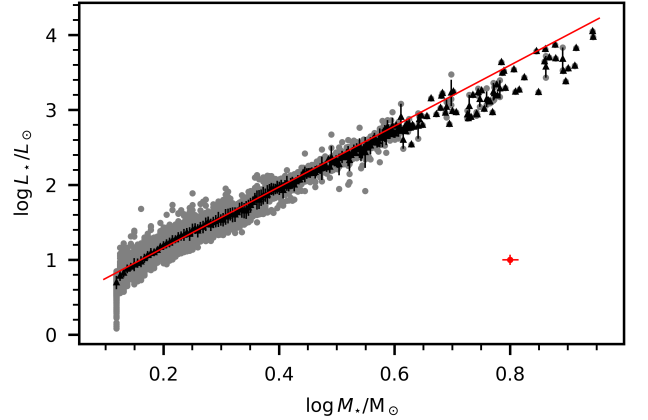


Fig. 6: Relation between the inferred stellar mass and the measured luminosity from Gaia of the stars in the sample. The solid red line indicates the fitted mass-luminosity relation, yielding $\log(L_{\star}/L_{\odot}) = 4.0623(6) \log(M_{\star}/M_{\odot}) + 0.3468(2)$. The black triangles indicate bin averaged-values. Average uncertainties are indicated by the red data point.

rotating g-mode pulsators with dominant amplitudes of a few mmag in that part of the HRD of a young open cluster, while Degroote et al. (2009) detected many candidate pulsators in that region from CoRoT space photometry as well. Along with the summary by Balona (2024) it seems firmly established now that g-mode pulsations occur all along the main sequence. Our results on the mass distribution are in line with these earlier observational studies.

The distribution of masses within the γ Dor range has a similar shape compared to the mass distribution derived by Mombarg et al. (2024) for the 539 genuine γ Dor stars observed by the *Kepler* mission. We do, however, find a peak in the number of stars around $1.4 M_{\odot}$ (top-left panel in Fig. 4), which is slightly lower in mass than the peak around $\sim 1.5 M_{\odot}$ observed for the *Kepler* prograde dipole g-mode pulsators considered by Mombarg et al. (2024). Furthermore, we observe a clear relation between the inferred mass and the observed luminosity,

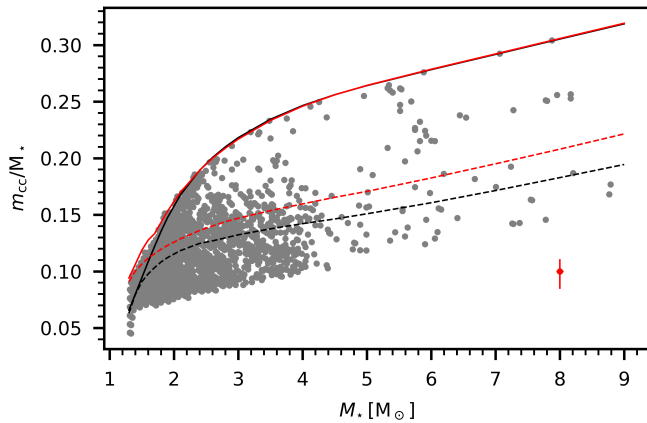


Fig. 7: Relation between the inferred stellar mass and fractional convective core mass of the stars in the sample. The solid black (red) line indicates the maximum core mass possible during the main sequence for $f_{\text{CBM}} = 0.005$ (0.025). The dashed lines show the maximum core mass at $X_c/X_{\text{ini}} = 0.4$ (same colour coding). Average uncertainties are indicated by the red data point.

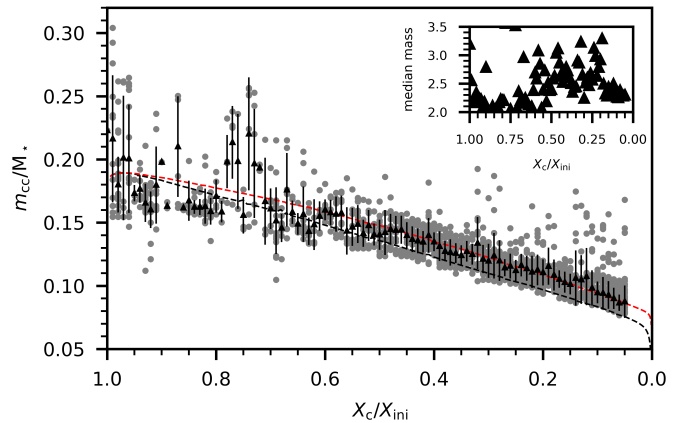


Fig. 9: Relation between the inferred fraction X_c/X_{ini} and the convective core mass for stars $M_* > 2M_\odot$. The black triangles indicate the average per bin and standard deviation. Two models are shown for $2.4 M_\odot$ and $\omega_0 = 0.05$, where the mass is representative of the median stellar mass per bin as shown in the inset. The black colour corresponds to $f_{\text{CBM}} = 0.005$, the red colour to $f_{\text{CBM}} = 0.025$.

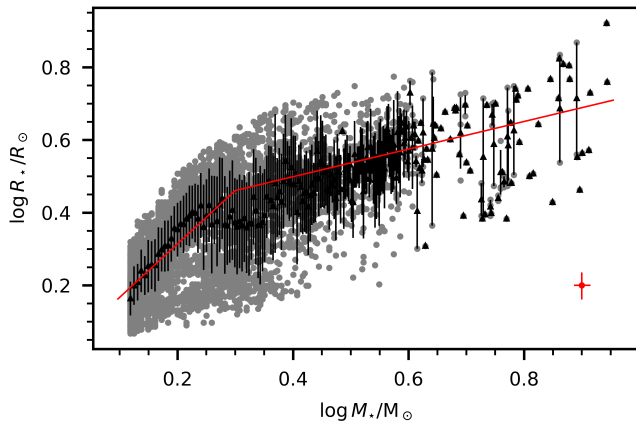


Fig. 8: Relation between the inferred stellar mass and the inferred stellar radius (from models) of the stars in the sample. The solid red line indicates a two-component linear fit to the data (see text). The black triangles indicate bin averaged-values. Average uncertainties are indicated by the red data point.

with $\log(L_*/L_\odot) = 4.0623(6) \log(M_*/M_\odot) + 0.3468(2)$ as shown in Fig. 6. This is a somewhat steeper mass–luminosity relation than the one found by Yakut et al. (2007) based on eclipsing binaries with a B-type star component, namely $\log(L_*/L_\odot) = 3.724 \log(M_*/M_\odot) + 0.162$. This is understood in terms of the increased importance of the radiation pressure with respect to the ideal gas law in the equation of state when going from F-type stars to B-type stars.

We find the majority of stars to form a plateau around 40 per cent of their initial hydrogen mass fraction left in the convective core, with a shape of a broad normal distribution around this peak value. The fact that most stars appear to be mid-main sequence is dominated by the γ Dor stars in the parameter region where convective flux blocking is active.

The bottom-left panel of Fig. 4 shows the distributions of the fractional convective core masses (m_{cc}/M_*). The peak originates from the fact that the sample contains both stars having

an initially growing convective core and stars with purely receding cores during the main sequence. We show the correlation between the mass of the convective core and total stellar mass in Fig. 7. The solid black and red lines show the maximum convective core mass as a function of total mass occurring in the grid, for CMB with $f_{\text{CBM}} = 0.005$ and $f_{\text{CBM}} = 0.025$, respectively. The dashed lines show the location for about half the main sequence, that is, at $X_c/X_{\text{ini}} = 0.4$. We see that numerous stars occur between the black and red lines, which reveals that a fraction of γ Dor stars with a mass below $\sim 1.8 M_\odot$ experience some level of CBM. However, most stars have a convective core mass below the maximum for modest CBM ($f_{\text{CBM}} = 0.005$), as was also found by Mombarg et al. (2024) for the genuine *Kepler* γ Dor sample. The maximum convective core mass of stars above roughly $2 M_\odot$ is not dependent on the CBM efficiency for the range of values we considered in our grids, but we recall that we excluded envelope mixing in our model grids while this may be an important ingredient to deduce the convective core mass of SPB stars (Pedersen et al. 2021; Pedersen 2022b). Asteroseismic modelling of oscillation frequencies is required to evaluate the cumulative effect of envelope mixing.

The bottom-right of Fig. 4 shows the distributions of the inferred stellar radii. Figure 8 shows the mass against the radius. As is common, we find more scatter in the mass–radius relation than in the mass–luminosity relation. We perform a fit with two linear components, where we place the knee at $\log M_*/M_\odot = 0.3$ ($2 M_\odot$), as stars below this mass are less inflated because they have a thin outer convective layer, while those with a higher mass have a fully radiative envelope. This yields $R_* \propto M_*^{1.469(3)}$ on the lower-mass end, and $R_* \propto M_*^{0.380(4)}$ on the higher-mass end, where the knee is located at $\log R_*/R_\odot = 0.4609(5)$.

In Fig. 9 (and Fig. A.3 for $Z = 0.014$), we show the derived correlation between the evolutionary stage (via X_c/X_{ini}) and the convective-core mass. In this figure, we only show stars above $2 M_\odot$, as the γ Dor stars are not equally sampled in mass for each bin in X_c/X_{ini} , since the more massive ones tend to be closer to the TAMS, while the less massive ones are closer to the ZAMS. The downward trend of the core mass over time

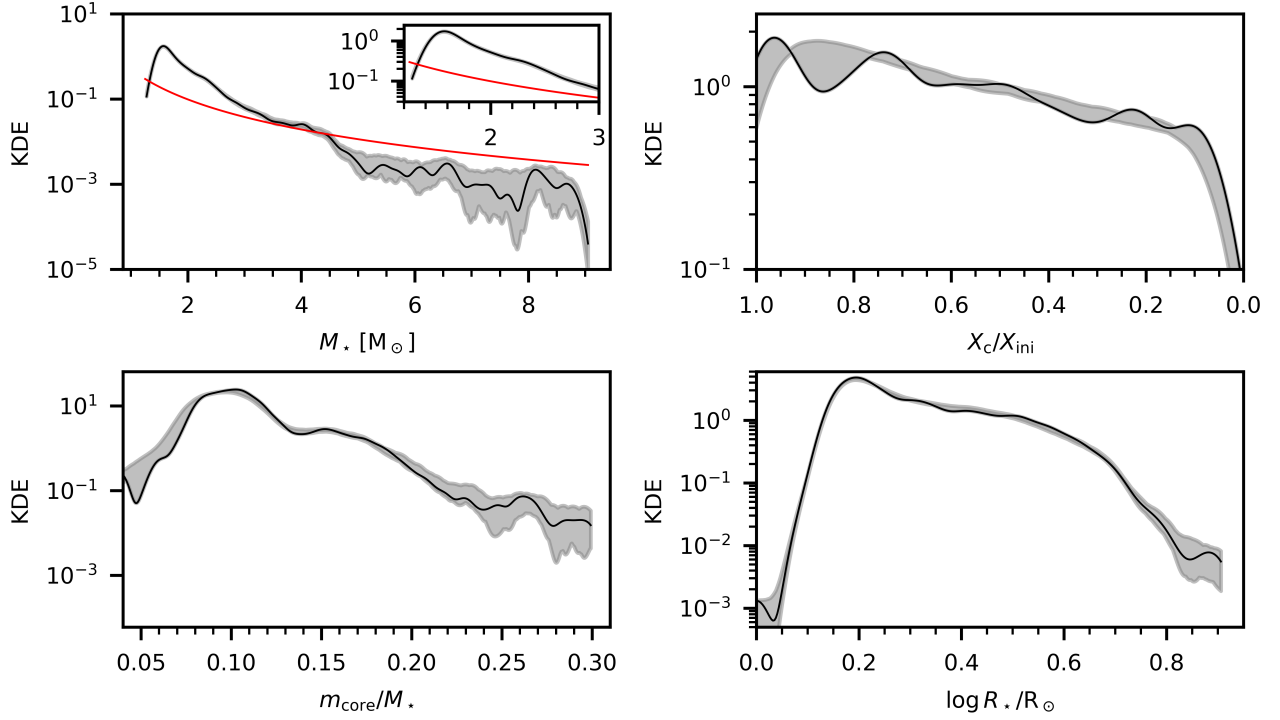


Fig. 10: Same as Fig. 4, but for $Z = 0.014$.

observed in Fig. 9 is consistent with a constant value of f_{CBM} along the main sequence, as can be seen from the models shown in this figure. However, variations in the value of f_{CBM} along the main sequence between the most extreme values used in our grids cannot be ruled out. While we do appreciate that the trend shown in Fig. 9 is to some extent enforced by the physics of the evolution models used, we note that this physics is based on detailed asteroseismic modelling studies (e.g. Pedersen et al. 2021; Michielsen et al. 2021; Burssens et al. 2023; Mombarg et al. 2024).

We repeat the exercise with a CNF trained on the grid with $Z = 0.014$. The results for the stars with $P_{\text{pred}} \geq 0.5$ are shown in Fig. 10. Our conclusions are qualitatively the same for this grid modelling keeping in mind a tight mass-metallicity relation, except that we now see an excess of stars that fall below the ZAMS line. Using this assumption of the metallicity, we obtain a mass-luminosity relation $\log(L_*/L_\odot) = 4.1825(5) \log(M_*/M_\odot) + 0.1057(2)$. For the mass-radius relation we find $R_* \propto M_*^{1.740(3)}$ on the lower-mass end, and $R_* \propto M_*^{0.623(3)}$ on the higher-mass end, where the knee is located at $\log R_*/R_\odot = 0.3945(3)$. The corresponding figures are shown in Appendix A. We expect most stars to have a true metallicity between the two extremes considered here. The more metal-poor models give better consistency for the observed pulsators as a population above the ZAMS based on the *Gaia* effective temperatures and luminosities, while their membership to the thin disk of the Milky Way would advocate for a more solar composition (see de Laverny et al. 2024, for a detailed discussion). We quantify the effect of assuming an incorrect metallicity in Appendix B.

5. Conclusions

In this paper, we have made use of conditional normalising flows trained on two grids of rotating one-dimensional stellar evolution

models calibrated by asteroseismology of γ Dor stars. Using this method, we derived masses, evolutionary stages (on the main sequence), convective core masses, and radii of ~14,000 new galactic g-mode pulsators. These form an all-sky sample and were discovered from sparse *Gaia* DR3 light curves. All stars were confirmed to be multiperiodic pulsators from TESS nominal high-cadence space photometry (Hey & Aerts 2024). This sample of TESS-confirmed g-mode main-sequence pulsators offers an increase with more than a factor 23 compared to the *Kepler* sample of asteroseismically modelled main-sequence g-mode pulsators.

The parameter determination we provide here is based on the stars' effective temperature and luminosity only, and hence represents evolutionary parameters. These estimates cannot be compared with parameters deduced from detailed asteroseismic modelling, but they offer initial estimates from the homogeneously treated *Gaia* DR3 data. The two observables for each star were coupled to a grid of stellar evolution models constructed with input physics deduced from the sample of 611 asteroseismically calibrated *Kepler* γ Dor pulsators covering rotation rates from almost zero to almost the critical rate. The homogeneously determined (convective core) masses and radii offer a valuable starting point for future ensemble asteroseismic modelling of these g-mode pulsators, once a sufficient number of their modes can be identified from high-cadence TESS or future PLATO (Rauer et al. 2024) space photometry.

Biases in the effective temperature of GSP-Phot used in this work might be present. The study by Avdeeva et al. (2024) report systematically higher effective temperatures measured by GSP-Phot compared those of the APOGEE survey. The largest discrepancies between GSP-Phot and APOGEE are, however, reported for $BP - RP$ values between 2 and 4.5 mag, whereas the stars in our sample have $BP - RP < 1.5$. A potential systematic overestimation of the effective temperature by GSP-Phot would